

Gurugram University Scheme of Studies and Examination
Bachelor of Technology(Electronics & Communication Engineering)

Semester IV

S. No	Category	Course Code	Course Title	Hours per week			Credits	Marks for Sessional	Marks for End Term Examination	Total Marks
				L	T	P				
1	PCC		Internet of Things	3	0	0	3	30	70	100
2	PCC		Digital & Data Communication	3	0	0	3	30	70	100
3	PCC		Electronic Measurement & Instrumentation	3	0	0	3	30	70	100
4	BSC		Probability Theory and Stochastic Processes	3	0	0	3	30	70	100
5	PCC		Computer Organization & Architecture	3	0	0	3	30	70	100
	PCC		Microprocessor & Interfacing	3	0	0	3	30	70	100
7	LC		Internet of Things Lab	0	0	2	1	50	50	100
8	LC		Electronic Measurement & Instrumentation Lab	0	0	2	1	50	50	100
9	LC		Microprocessor & Interfacing Lab	0	0	2	1	50	50	100
10	LC		PCB Design Lab	0	0	2	1	50	50	100
11	MC		Scientific & Technical writing Skills*	2	0	0	-	30	70	100*
Total				28			22	380	620	1000

NOTE:

1. Scientific & Technical writing Skills*: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

2. At the end of the 4th semester, each student has to undergo Practical Training of 4/6 weeks in an Industry/ Institute/ Professional Organization/ Research Laboratory/ trainingcenter etc. and submit the typed report along with a certificate from the organization & its evaluation shall be carried out in the 5th Semester.

INTERNET OF THINGS

Course code					
Category	Professional Core Course				
Course title	Internet of Things				
Scheme and Credits	L	T	P	Credits	Semester :IV
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES: The objective of this course is to impart necessary and practical knowledge of components of Internet of Things and develop skills required to build real-life IoT based projects.

UNIT I

Introduction to IoT: Architectural Overview, Design principles and needed capabilities, IoT Applications, Sensing, Actuation, Basics of Networking, M2M and IoT Technology Fundamentals- Devices and gateways, Data management, Business processes in IoT, Everything as a Service(XaaS), Role of Cloud in IoT, Security aspects in IoT.

UNIT II

Elements of IoT: Hardware Components- Computing (Arduino, Raspberry Pi), Communication, Sensing, Actuation, I/O interfaces. Software Components- Programming API's (using Python/ Node.js/ Arduino) for Communication. Protocols-MQTT, ZigBee, Bluetooth, CoAP, UDP, TCP.

UNIT III

IoT Application Development: Solution framework for IoT applications- Implementation of Device integration, Data acquisition and integration, Device data storage- Unstructured data storage on cloud/local server, Authentication, authorization of devices.

UNIT IV

IoT Case Studies : IoT case studies and mini projects based on Industrial automation, Transportation, Agriculture, Healthcare, Home Automation

COURSE OUTCOMES: After the completion of this course, the students will be able to:

1. Understand internet of Things and its hardware and software components
2. Interface I/O devices, sensors & communication modules.
3. Understand various IoT protocols.
4. Analyse data processing in IoT.
5. Remotely monitor data and control devices
6. Develop real life IoT based projects

1. Vijay Madiseti, Arshdeep Bahga, 'Internet of Things, "A Hands on Approach", University Press
2. Dr. SRN Reddy, Rachit Thukral and Manasi Mishra, "Introduction to Internet of Things: A practical Approach", ETI Labs
3. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press
4. Jeeva Jose, "Internet of Things", Khanna Publishing House, Delhi
5. Adrian McEwen, "Designing the Internet of Things", Wiley
6. Raj Kamal, "Internet of Things: Architecture and Design", McGraw Hill
7. Cuno Pfister, "Getting Started with the Internet of Things", O Reilly Media

DIGITAL & DATA COMMUNICATION

Course Code					
Category	Professional Core Courses				
Course title	Digital & Data Communication				
Scheme	L	T	P	Credits	Semester : IV
	3	1	0	3	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. Understand the working principles of Data Communication.
2. Understand the Data link layer.
3. Understand the network security.

UNIT I

Data Transmission Basics: Communication model - Simplex, Half duplex, Full duplex transmission. Periodic analog signals - Sine wave, Amplitude, Phase, Wavelength, Time and frequency domain, Bandwidth. Analog & digital data and signals. Transmission impairments - Attenuation, Delay distortion, Noise. Data rate limits - Noiseless channel, Nyquist bandwidth, Noisy channel, Shannon's capacity formula.

UNIT II

Transmission Media: Guided transmission media - Twisted pair, Coaxial cable, Optical fiber. Unguided media - Radio waves, Terrestrial microwave, Satellite microwave, Infrared. Wireless propagation - Ground wave propagation, Sky wave propagation, Line-of-Sight (LoS) propagation.

Digital Transmission and Analog Transmission: Digital data to digital signal – Non-Return-to-Zero (NRZ), Return-to-Zero (RZ), Multilevel binary, Biphasic. Analog data to digital signal - Sampling theorem, Pulse Code Modulation (PCM), Delta Modulation (DM). Digital data to analog signal - Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), Phase Shift Keying (PSK). Analog data to analog signal - Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM).

UNIT III

Multiplexing and Spread Spectrum: Multiplexing - Frequency Division Multiplexing (FDM), Wavelength Division Multiplexing (WDM), Time Division Multiplexing (TDM), Characteristics, Synchronous TDM, Statistical TDM. Spread spectrum techniques - Direct Sequence Spread Spectrum (DSSS), Frequency Hopping Spread Spectrum (FHSS), Code Division Multiplexing, Code Division Multiple Access (CDMA).

UNIT IV

Error Detection, Correction and Switching: Digital data communication techniques - Asynchronous transmission, Synchronous transmission. Detecting and correcting errors - Types of errors, Parity check, Checksum, Cyclic Redundancy Check (CRC), Forward Error Correction (FEC), Hamming distance, Hamming code. Basic principles of switching - Circuit switching, Packet switching, Message switching.

Course Outcomes

1. Identify the characteristics of signals for analog and digital transmissions
2. Identify the issues in data transmission .
3. Select transmission media based on characteristics and propagation modes
4. Choose appropriate signal encoding techniques for a given scenario
5. Illustrate multiplexing and spread spectrum technologies
6. Use error detection, correction and switching techniques in data communication

Text/Reference Books:

1. Forouzan B. A., Data Communications and Networking, 5/e, McGraw Hill, 2013.
2. William Stallings, Data and Computer Communication 9/e, Pearson Education, Inc. 273 COMPUTER SCIENCE AND ENGINEERING

References Books:

1. Schiller J., Mobile Communications, 2/e, Pearson Education, 2009.

ELECTRONIC MEASUREMENT AND INSTRUMENTATION

Course Code						
Category	Professional Core Courses					
Course title	Electronic measurement and Instrumentation					
Scheme	L	T	P	Credits	Semester : IV	
	3	1	0	3		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	03 Hrs					

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To introduce the fundamentals of Electronics Instruments and Measurement providing an in-depth understanding of Measurement errors.
2. Digital Storage Oscilloscope, Function Generator and Analyzer, Display devices, Data acquisition systems and transducers.
3. To address the underlying concepts and methods behind Electronics measurements.

UNIT I

OSCILLOSCOPE:Block diagram, study of various stages in brief, high frequency CRO considerations. Sampling and storage oscilloscope.

GENERATION & ANALYSIS OF WAVEFORMS:Block diagram of pulse generators, signal generators, function generators wave analysers, distortion analysers, spectrum analyser, Harmonic analyser, introduction to power analyser.

UNIT II

ELECTRONIC INSTRUMENTS:Instruments for measurement of voltage, current & other circuit parameters, Q meters, R.F. Power measurements, introduction to digital meters.

FREQUENCY & TIME MEASUREMENT:Study of decade counting Assembly(DCA), frequency measurements, period measurements, Universal counter, Introduction to digital meters.

UNIT III

DISPLAY DEVICES:Nixie tubes, LED's LCD's, discharge devices.

TRANSDUCERS:Classification, Transducers of types: RLC photocell, thermocouples etc. basic schemes of measurement of displacement, velocity, acceleration, strain, pressure, liquid level & temperature.

UNIT IV

INTRODUCTION TO SIGNAL CONDITIONING:

DC signal conditioning system, AC signal conditioning system, data acquisition and conversion system

Course Outcome:

1. Analyze the performance characteristics of each instrument
2. Illustrate basic meters such as voltmeters and ammeters.
3. Explain about different types of signal analyzers.
4. Explain the basic features of oscilloscope and different types of oscilloscopes
5. Identify the various parameters that are measurable in electronic instrumentation.
6. Employ appropriate instruments to measure given sets of parameters.

TEXT / REFERENCE BOOKS.

1. A course in Electrical & Electronics Measurements & Instrumentation : A.K.Sawhney; Dhanpat Rai Sons.
2. Electronics Instrumentation & Measurement Techniques : Cooper; PHI.

PROBABILITY THEORY AND STOCHASTIC PROCESSES

Code					
Category	Basic Science Courses				
Course title	Probability Theory and Stochastic Processes				
Scheme	L	T	P	Credits	Semester : IV
	3	0	0	3	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To introduce the fundamentals of probability theory and random processes and illustrate these concepts with engineering applications.
2. To introduce random variables.
3. The course introduces the concept of Stochastic Processes.
4. To understand regression analysis.

UNIT I

Sets and set operations; Probability space; Conditional probability and Bayes theorem; Combinatorial probability and sampling models. Discrete random variables, probability mass function, probability distribution function, example random variables and distributions; Continuous random variables, probability density function, probability distribution function, example distributions

UNIT II

Joint distributions, functions of one and two random variables, moments of random variables; Conditional distribution, densities and moments; Characteristic functions of a random variable; Markov, Chebyshev and Chernoff bounds;

UNIT III

Random sequences and modes of convergence (everywhere, almost everywhere, probability, distribution and mean square); Limit theorems; Strong and weak laws of large numbers, central limit theorem. Random process. Stationary processes. Mean and covariance functions. Ergodicity. Transmission of random process through LTI. Power spectral density.

UNIT IV

Regression analysis (linear and non-linear), Confidence intervals, Hypothesis testing, Error analysis

Course Outcomes:

1. Develop understanding of basics of probability theory.
2. Understand random variables.
3. Identify different distribution functions and their relevance.
4. Apply the concepts of probability theory to different problems.
5. Extract parameters of a stochastic process and use them for process characterization.
6. Apply regression analysis.

Text/Reference Books:

1. H. Stark and J. Woods, 'Probability and Random Processes with Applications to Signal Processing,' Third Edition, Pearson Education
2. A. Papoulis and S. Unnikrishnan Pillai, 'Probability, Random Variables and Stochastic Processes,' Fourth Edition, McGraw Hill.
3. K. L. Chung, Introduction to Probability Theory with Stochastic Processes, Springer International
4. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability, UBS Publishers,
5. S. Ross, Introduction to Stochastic Models, Harcourt Asia, Academic Press

COMPUTER ORGANIZATION & ARCHITECTURE

Course Code					
Category	Professional Core Courses				
Course title	Computer organization & architecture				
Scheme	L	T	P	Credits	Semester : IV
	3	0	0	3	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. Discuss the basic concepts and structure of computers.
2. Understand concepts of register transfer logic and arithmetic operations.
3. Explain different types of addressing modes and memory organization.
4. Summarize the Instruction execution stages.

UNIT-I

Basics of a computer system: Evolution, Ideas, Technology, Performance, Power wall, Uniprocessors to Multiprocessors. Addressing and addressing modes. Instructions: Operations and Operands, Representing instructions, Logical operations, control operations. Case study - instruction sets of some common CPUs. Fixed point Addition, Subtraction, Multiplication and Division. Floating Point arithmetic, High performance arithmetic, Subword parallelism.

UNIT-II

Introduction, Logic Design Conventions, Building a Datapath – A Simple Implementation scheme -An Overview of Pipelining – Pipelined Datapath and Control. Data Hazards: Forwarding versus Stalling, Control Hazards, Exceptions, Parallelism via Instructions.

UNIT-III

MEMORY AND I/O ORGANIZATION:

Memory hierarchy, Memory Chip Organization, Cache memory, Virtual memory. Parallel Bus Architectures, Internal Communication Methodologies, Serial Bus Architectures, Mass storage, Input and Output Devices.

UNIT-IV

Parallel processing architectures and challenges, Hardware multithreading, Multicore and shared memory multiprocessors, Introduction to Graphics Processing Units, Clusters and Warehouse scale computers – Introduction to Multiprocessor network topologies.

Course Outcomes: At the end of this course, students will demonstrate the ability to:

1. Understand basics of a computer system.
2. Understanding Logic gates, flip flops and counter
3. Clear Understanding of Computer Architecture
4. Pipeline processing
5. RISC and CISC architectures
6. Develop a base for advance micro-processors Reference

Text/ reference books:

1. Computer System Architecture: By M. Morris Mano.
2. Structured Computer Organization: By Tanenbaum.
3. Computer Organization: By Stallings.
4. Computer Architecture and Organization: By Hayes.
5. Microprocessor Architecture, Programming, and Applications with the 8085 Ramesh S. Gaonkar Pub: Penram International

MICROPROCESSORS AND INTERFACING

Course Code					
Category	Professional Core Courses				
Course title	Microprocessors and Interfacing				
Scheme	L	T	P	Credits	Semester : IV
	3	1	0	3	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course.

1. To develop an in-depth understanding of the operation of microprocessors.
2. To master the assembly language programming using concepts like assembler directives, procedures, macros, software interrupts etc.
3. To create an exposure to basic peripherals, its programming and interfacing techniques.
4. To understand the concept of Interrupts and interfacing details of 8086 .

UNIT-I

THE 8086 MICROPROCESSOR ARCHITECTURE: Architecture, block diagram of 8086, details of sub-blocks such as EU, BIU; memory segmentation and physical address computations, program relocation, addressing modes, instruction formats, pin diagram and description of various signals.

UNIT II

INSTRUCTION SET OF 8086: Instruction execution timing, assembler instruction format, data transfer instructions, arithmetic instructions, branch instructions, looping instructions, NOP and HLT instructions, flag manipulation instructions, logical instructions, shift and rotate instructions, directives and operators, programming examples.

UNIT III

Concepts of virtual memory, Cache memory, Architecture & Instructions set of X86 family Microprocessors (80186, 80286, 80386, 80486). Enhanced features of Pentium, Pentium Pro, Pentium-II, Pentium-III, Pentium-IV, Multi-core Technology, Mobile Processor.

UNIT IV

INTERFACING DEVICE: Serial I/O, parallel I/O, A/D & D/A converters. 8255 Programmable peripheral interface, interfacing keyboard and seven segment display, 8254 (8253) programmable interval timer, 8259A programmable interrupt controller, Direct Memory Access and 8237 DMA controller.

Course Outcomes: At the end of this course students will be able to:

1. Understand the fundamentals of Microprocessors.
2. Understand the internal design of 8086 microprocessor along with the features .
3. Analyze a detailed s/w & h/w structure of the Microprocessor
4. Illustrate how the different peripherals (8086) are interfaced with Microprocessor.
5. Analyze the programming. of Microprocessors
6. Evaluate the data transfer information through serial & parallel ports.

Text / References Books:

1. Douglas Hall, "Microprocessor & Interfacing", 2nd Edition, TMH, 2006.
2. Muhammad A. Mazidi, "The 8051 Microcontroller And Embedded Systems Using Assembly and C", 2nd Edition., PHI, 2012.
3. Text / Reference Books: 1. D. V. Hall, Microprocessors and interfacing, Tata McGraw-Hill, 2nd Edition, 2006.
4. Ray A. K. and Burchandi, Advanced Microprocessors and Peripherals Architectures, Programming and Interfacing, Tata McGraw Hill, 2002.
5. Brey, The Intel Microprocessors 8086- Pentium Processor, 8th Edition, Pearson Education.
6. M. A. Mazidi, J. P. Maizidi and Danny Causey, The X86 PC: Assembly Language, Design and interfacing, 5th Edition, Pearson Education, 2017.
7. Liu Yu-Chang and Gibson Glenn A., Microcomputer Systems: The 8086/8088 Family: Architecture, Programming and Design, 2nd Edition, Pearson Education, 2015.
8. L. B. Das, The X86 Microprocessor (Architecture, Programming and Interfacing), 2nd Edition, Pearson Education, 2014.
9. Daniel Tabak, Advanced Microprocessor", Tata McGraw-Hill, 2nd Edition, 2012.
10. B. Ram, Fundamentals of Microprocessor and Microcomputers, Dhanpat Rai Publications, 5th edition, 2008.

11. MICROPROCESSOR & INTERFACING LAB

Course code					
Category	Laboratory Courses				
Course title	Microprocessor & Interfacing Lab				
Scheme and Credits	L	T	P	Credits	Semester : IV
	0	0	2	2	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

Note:

1. Total ten experiments are to be performed in the semester.
2. At least seven experiments should be performed from the above list. Remaining three experiments
3. Should be performed as designed and set by the concerned institution as per the scope of the syllabus.

Course Objectives: The objectives of this course are as under:

1. To introduce the students with 8086 kit.
2. To acquaint them to do assembly language programming of 8086.
3. To acquaint them to do assembly language programming of 8086 for interfacing of peripherals.

LIST OF EXPERIMENTS:

1. To study the architecture of 8086 microprocessor and 8086 microprocessor kit.
2. Write a program to add the contents of the memory location to the content of other memory location and store the result in 3rd memory location.
3. Write a program to add 16 bit number using 8086 instruction set.
4. Write a multiplication of two 16 bit numbers using 8086 instruction set.
5. Write a program for division of two 16 bit numbers using 8086 instruction set.
6. Write a program factorial of a number.
7. Write a Program to transfer a block of data with & without overlap.
8. Write a program to find the average of two numbers.
9. Write a Program to check whether data byte is odd or even
10. Write a program to find maximum number in the array of 10 numbers.
11. Write a program to find the sum of the first 'n' integers.
12. Write a program to generate a square wave.
13. Write a program to generate a rectangular wave.
14. Write a program to generate a triangular wave.

Course Outcomes: At the end of the course, students will demonstrate the ability to:

1. Understand the fundamentals of Microprocessors.
2. Do assembly language programming of 8086.
3. Do assembly language programming of 8086 for interfacing of peripherals.
4. Can generate different kind of wave forms using assembly language.

ELECTRONIC MEASUREMENT & INSTRUMENTATION LABORATORY

Course Code					
Category	Laboratory Courses				
Course title	Electronic Measurement & Instrumentation Laboratory				
Scheme	L	T	P	Credits	Semester : IV
	0	0	2	1	
Class Work	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02Hrs				

Notes:

- (i) At least 10 experiments are to be performed by students in the semester.
- (ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
- (iii) Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS

1. Study blocks wise construction of a analog oscilloscope & Function generator.
2. Study blocks wise construction of a Multimeter & frequency counter.
3. Study Measurement of different components & parameters like Q of a coil etc using LCRQ meter.
4. Study of distortion factor meter and determination of the % distortion of the given oscillator
5. Determine output characteristics of a LVDT and Measure displacement using LVDT
6. Study characteristics of temperature transducer like Thermocouple, Thermistor & RTD with implementation of a small project using signal conditioning circuits like instrumentation amplifier.
7. Measurement of Strain using Strain Gauge.
8. To study differential pressure transducer & signal conditioning of output signal.
9. Measurement of level using capacitive transducer.
10. Study of Distance measurement using ultrasonic transducer.

Lab Outcome: At the end of this lab, students will demonstrate the ability to;

1. Analyze the performance characteristics of each instrument
2. Illustrate basic meters such as voltmeters and ammeters.
3. Explain about different types of signal analyzers.
4. Explain the basic features of oscilloscope and different types of oscilloscopes
5. Identify the various parameters that are measurable in electronic instrumentation.
6. Employ appropriate instruments to measure given sets of parameters.

INTERNET OF THINGS LAB

Course Code						
Category	Laboratory Courses					
Course title	Internet of Things LAB					
Scheme	L	T	P	Credits	Semester: IV	
	0	0	2	1		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	02 Hrs					

Notes:

1. At least 10 experiments are to be performed by students in the semester.
2. At least 7 experiments should be performed from the list, remaining three experiments may either be Performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
3. Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS

1. To study the architecture and pins of Arduino
2. Write a program for blinking of LED and vary its intensity.
3. To sense the available networks using Arduino.
4. Measure the distance using ultrasonic sensor.
5. To detect the vibration of an object .
6. Sense the temperature and display it on LCD display.
7. Vary the intensity of light of LED using LDR.
8. MySQL Queries Database installation in Raspberry Pi
9. SQL Queries by Fetching data from Raspberry Pi.
10. Switch light ON and OFF based on the input of user using Raspberry Pi.

Lab Outcome: At the end of this lab, students will demonstrate the ability to;

1. Understand the basics of IoT.
2. Learn Architecture and enabling technologies
3. Implement application of different sensors
4. Can design project using Arduino and Raspberry Pi

PCB & WORKSHOP LABORATORY

Course code					
Category	Laboratory Courses				
Course title	PCB & WORKSHOP LAB				
Scheme and Credits	L	T	P	Credits	Semester : IV
	0	0	2	2	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

Course Objective: To create interest in Hardware Technology.

1. Winding shop: Step down transformer winding of less than 5VA.
2. Soldering shop: Fabrication of DC regulated power supply
3. PCB Lab: (a) Artwork & printing of a simple PCB.
(b) Etching & drilling of PCB.
4. Wiring & fitting shop: Fitting of power supply along with a meter in cabinet.
5. Testing of regulated power supply fabricated.

Experiment to be performed

1. Introduction & Hands on experience to use circuit creation & simulation software like TINAPRO , MULTISIM, PSPICE or ORCAD etc.
2. Design a full wave centre tapped rectifier & study the effect of capacitive filter & its output on a virtual oscilloscope.
3. Design a RLC resonance circuit & verify the transient & phase response for different values of R,L &C.
4. Design a circuit for a fixed power supply.
5. Design a half adder using discrete components & verify the timing diagrams.
6. Convert the power supply circuit into PCB & simulates its 2D & 3D view.
7. PCB printing using screen printing or any other technique.
8. Etching of the above PCB.
9. UV exposure & Drilling of PCB.
10. Coating of etched PCB to protect it from oxidation.
11. Fabrication & placing of components as per above power supply circuit.
12. Testing of above circuit.

Course Outcomes: At the end of the course, students will be able to:

1. Understand the characteristics of diodes and filter circuits.
2. Understand the operation and characteristics of different types of rectifiers.
3. Understand the operation and characteristics of power supply.

SCIENTIFIC & TECHNICAL WRITING SKILLS

Course Code						
Category	Mandatory Course					
Course title	Scientific & Technical writing Skills					
Scheme	L	T	P	Credits	Semester: IV	
	2	0	0	0		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	3Hrs					

The following course content to conduct the activities is prescribed for the Scientific & Technical writing Skills Lab:

1. Activities on Writing Skills - Structure and presentation of different types of writing - letter writing/ Resume writing/ e-correspondence/ Technical report writing/ Portfolio writing - planning for writing - improving one's writing.
2. Activities on Presentation Skills - Oral presentations (individual and group) through JAM sessions/seminars/PPTs and written presentations through posters/ projects/ reports/ e-mails/ assignments etc.
3. Activities on Group Discussion and Interview Skills - Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation- Concept and process, pre-interview planning, opening strategies, answering strategies, interview through tele-conference & video-conferencing and Mock Interviews.

Text references:

1. A Course Book of Advanced Communication Skills (ACS) Lab published by Universities Press, Hyderabad.

Books Recommended:

1. Technical Communication by Meenakshi Raman & Sangeeta Sharma, Oxford University Press 2009.
2. Advanced Communication Skills Laboratory Manual by Sudha Rani, D, Pearson Education 2011.
3. Technical Communication by Paul V. Anderson , 2007. Cengage Learning pvt. Ltd. New Delhi.
4. Business and Professional Communication: Keys for Workplace Excellence, Kelly M. Quintanilla & Shawn T. Wahl. Sage South Asia Edition. Sage Publications, 2011.
5. The Basics of Communication: A Relational Prespective, Stev Duck & David T. Mc Mahan. Sage South Asia Edition. Sage Publications, 2012.
6. English Vocabulary in Use series, Cambridge University Press 2008.
7. Management Shapers Series by Universities Press (India) Pvt Ltd., Himayatnagar, Hyderabad 2008.
8. Handbook for Technical Communication by David A. McMurrey & Joanne Buckley, 2012. Cengage Learning.
9. Communication Skills by Leena Sen, PHI Learning Pvt Ltd., New Delhi, 2009.
10. Handbook for Technical Writing by David A McMurrey & Joanne Buckely CENGAGE Learning 2008.
11. Job Hunting by Colm Downes, Cambridge University Press 2008.
12. Master Public Speaking by Anne Nicholls, JAICO Publishing House, 2006.
13. English for Technical Communication for Engineering Students, Aysha Vishwamohan, Tata Mc graw Hill 2009.
14. Books on TOFEL/ GRE/ GMAT/ CAT/ IELTS by Barron's/ DELTA/ Cambridge University Press.
15. International English for Call Contres by Barry Tomalin and Suhashini Thomas, Macmillan Publishers, 2009.